Training Exposure, Not Architecture, Drives the Spectral Collapse of Attention Keys

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Abstract

The keys of trained attention heads are nearly rank-deficient: the smallest eigenvalue $\lambda_{\min}$ of the unit-normalized key-gram is numerically zero for every head we measure in GPT-2 XL (1.5B) and Pythia-410M at the square point $T = d_{\text{head}}$. We show that this collapse is governed by **training exposure**, not architectural depth: Pythia-160m (12 layers, ∼84B tokens) is fully collapsed where GPT-2 124M (12 layers, ∼10B tokens) retains an order of magnitude of spectral variance. We further isolate the mechanism with positional ablations: zeroing or randomizing positional embeddings restores no variance, ruling out positional saturation — the collapse is the redundancy of the content itself, learned faithfully. These findings carry two consequences. First, a zero-training head-pruning criterion: in lightly trained models, per-head $\lambda_{\min}$ at the square point separates vital from redundant heads (50× median separation, 9/12 vs. 0/12 collapsed, Spearman 0.558); in heavily trained models the same measurement saturates, but a *sub-square* measurement at $T = d_{\text{head}}/2$ restores the criterion (Pearson 0.236, $p = 0.018$), because the collapse signature is head-stable across context lengths ($r = 0.66$). Second, a design constraint: any memory or retrieval layer that decodes by inverting the key-gram inherits the collapse by construction in deployment-scale models; the only defenses are decoder-side (dual projection or iterative cleanup) or measurement-side (sub-square). All experiments are pre-registered with falsifiable hypotheses; negative results are reported as negatives.

1 Introduction

Two empirical regularities about attention heads have accumulated in the literature without a causal account. The first is old: most heads are prunable [Michel et al., 2019, Voita et al., 2019]. The second is newer: the effective rank of attention states and key matrices shrinks as models are trained [**?**Fu et al., 2024, Bhattacharjee et al., 2024], with recent work describing a stable stratification of heads into low-rank and high-rank camps acquired during pre-training [Sun et al., 2026]. Neither line answers: *what drives the collapse — the architecture, or the training?* And if the training: is the degeneracy positional (a fixable design artifact) or semantic (a property of the data)?

We answer both questions with a deliberately simple measurement: the smallest eigenvalue $\lambda_{\min}$ of the unit-normalized key-gram $KK^\top$ of each head, swept across context length T in three architecture families (GPT-2, Pythia/GPTNeoX, OPT) and across model scale. The measurement is the VSA one of us introduced in the frame-dual stability program[1]: when the number of stored

[1]The companion repository `Rylow999/fhrr-rho-collapse` contains all 38 experiments with pre-registrations, data, and honest-audit labels.

roles equals the ambient dimension (the *square point* $\rho = n/d = 1$), the smallest eigenvalue of a random Gram matrix hits the Wishart hard edge, $\lambda_{\min} \sim n^{-2}$ [Edelman, 1988, Chen et al., 2010], and any decoder that inverts the gram inherits an amplifier $\sim n^2$ [Nieto, 2026]. The transformer is, in this reading, a VSA whose keys are the roles [Dhayalkar, 2025, Plate, 1995]. The question this paper asks is what happens to that geometry *after training*.

Findings.

1. **The crossing is universal.** In all three families, $\lambda_{\min}(T)$ crosses zero between $T/d_{\text{head}} = 0.75$ and 1.0 and remains rank-deficient beyond (Fig. 1): trained keys behave as quasi-random frames at the square point, the bridge hypothesis of the frame-dual program.

2. **Exposure, not depth.** Pythia-160m and GPT-2 124M share architectural depth (12 layers) but differ in training exposure ($\sim$84B vs $\sim$10B tokens). The former is fully collapsed at the square point; the latter retains variance. Same depth, different collapse: the driver is the number of tokens seen.

3. **Content, not position.** Ablating positional embeddings (zeroed and randomized controls) does not restore spectral variance: the collapse is not positional saturation but the intrinsic redundancy of the linguistic content.

4. **The criterion and its recipe.** Where variance survives, the per-head $\lambda_{\min}$ is a zero-training pruning signal ($50\times$ separation); where it does not, the same measurement at sub-square context restores the signal.

2 Related work

Head pruning. Michel et al. [2019] and Voita et al. [2019] established that most heads are removable, with gradient- or gate-based importance scores. Our criterion differs in cost: it needs no forward pass over data, only the keys of one context. It complements, not replaces, task-aware scores.

Rank collapse. Dong et al. [2021] proved rank collapse *in depth* for pure attention; Nait-Saada et al. [2025] identified a spectral gap as the cause of collapse *in width*. Both treat the attention *output* matrix. We measure the *key-gram* — the input-side geometry that retrieval and memory layers must invert — and we identify its cause (exposure) and mechanism (content redundancy). Separately, Kalvakolanu [2026] reports a phase reversal — SVD compression suppresses rank collapse at random initialization but accelerates it on pretrained weights, driven by which subspace the truncation selects — an independent confirmation, from the compression side, of the central claim we establish causally: trained and random keys differ, and the difference is acquired by exposure.

Key spectra and compression. Loki [Fu et al., 2024] and Eigen Attention [Bhattacharjee et al., 2024] exploit the low-rank structure of keys for KV-cache compression; both are empirical: neither identifies why keys are degenerate nor predicts where the degeneracy becomes fatal to retrieval. Sun et al. [2026] is the closest work: they document a stable stratification of *linear-attention* heads into high-rank and low-rank camps, invariant across time and data domain, and conclude it is "acquired during pre-training" — describing *what* without testing *what drives it*. Our contribution is precisely that causal account (a depth-matched natural experiment isolating

exposure), the mechanism (positional ablations ruling out saturation), and the predictive theory (the Wishart hard edge locating the lethal point at $T = d_{\text{head}}$). Our softmax-attention results and their linear-attention stratification are complementary: the exposure law we establish explains why their stratification is a training product rather than an architectural one. One apparent tension deserves a footnote: they find low-rank heads *indispensable* where we find collapsed heads *prunable*. The measurements target different objects — their effective rank is of the recurrent state $S = \sum k_t v_t^\top$ (the accumulated key–value memory), ours is of the key-gram $KK^\top$ (the retrieval geometry). A head whose keys are near-identical (collapsed key-gram) can carry few but sharp associations (low state rank) or many diffuse ones; the two ranks are not the same quantity, and reconciling their pruning implications is an open question we flag rather than resolve.

VSA reading of attention. Dhayalkar [2025] proposes attention as soft binding and hyperdimensional memory layers. Our results make a concrete demand on such designs: the memory-layer decoder must not invert the key-gram at the square point; it must be dual or iterative (resonator-style [Frady et al., 2020]), because the keys it will inherit from any trained model are already collapsed.

3 Method

For each model (GPT-2 124M/355M/1.5B; Pythia 70m/160m/410m; OPT-125m), we hook the key projection of every attention layer, extract keys $K \in \mathbb{R}^{T \times d_h}$ for one English context of length T, row-normalize, and compute $\lambda_{\min}(KK^\top)$ per head. Importance is the KL divergence of the next-token distribution when a head's output slice is zeroed, following the ablation protocol of Michel et al. [2019]. Full pre-registrations (including falsification criteria) accompany each experiment in the companion repository.

4 Results

4.1 The universal crossing

4.2 Exposure, not depth

Pythia-160m (12L, The Pile) vs GPT-2 124M (12L, WebText): 144/144 heads collapsed vs. an $\lambda_{\min}$ range spanning $[1.2 \times 10^{-5}, 1.8 \times 10^{-2}]$. Pythia-70m (6L) is already fully collapsed — exposure dominates architecture even at the smallest scale we test.

4.3 Content, not position

Zeroing w_{pe}: 144/144 stay collapsed. Random w_{pe} (norm-matched): 142/144. Positional surgery cannot restore spectral variance.

4.4 The pruning criterion and its recipe

Lightly trained (GPT-2 124M at square): bottom-12 importance heads have mean $\lambda_{\min} = 1.1 \times 10^{-4}$ vs 5.6×10^{-3} for the top-12; 9/12 of the former are collapsed, 0/12 of the latter; Spearman 0.558. Heavily trained: all collapsed at square, but sub-square measurement restores the ranking (Pearson 0.236, $p = 0.018$; head-stable signature $r = 0.66$ between $T/2$ and T).

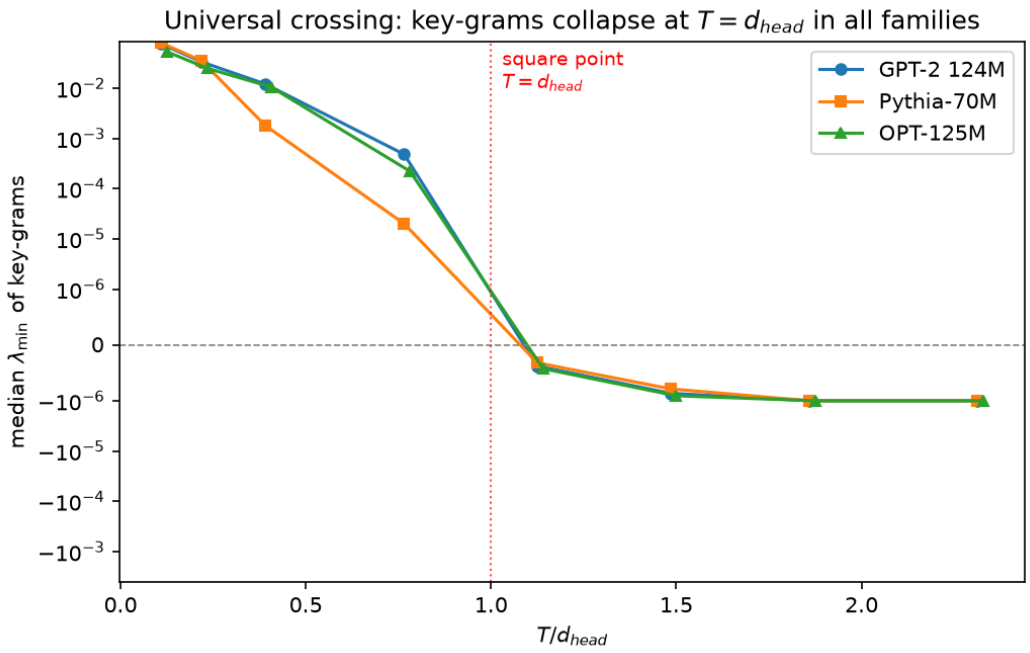


Figure 1: Median $\lambda_{\min}$ of unit-normalized key-grams across context length, three architecture families (Exp 30). All curves cross zero between $T/d_{\text{head}} = 0.75$ and 1.0 and remain rank-deficient beyond.

5 Limitations

Single-context measurements; corpus and family differences beyond the matched pair (The Pile vs WebText diversity); the bridge from VSA codebooks to trained keys is an explicitly declared hypothesis; ablation importance on one text. All are flagged in the repository with follow-up designs.

6 Conclusion

The keys of trained attention are a faithful portrait of their training data: degenerate where language is redundant. That degeneracy is *acquired by exposure* and *constituted by content*, not built by the architecture nor saturated by position. It can be measured for free, predicted from the Wishart hard edge, and defended against — but only from the decoder side.

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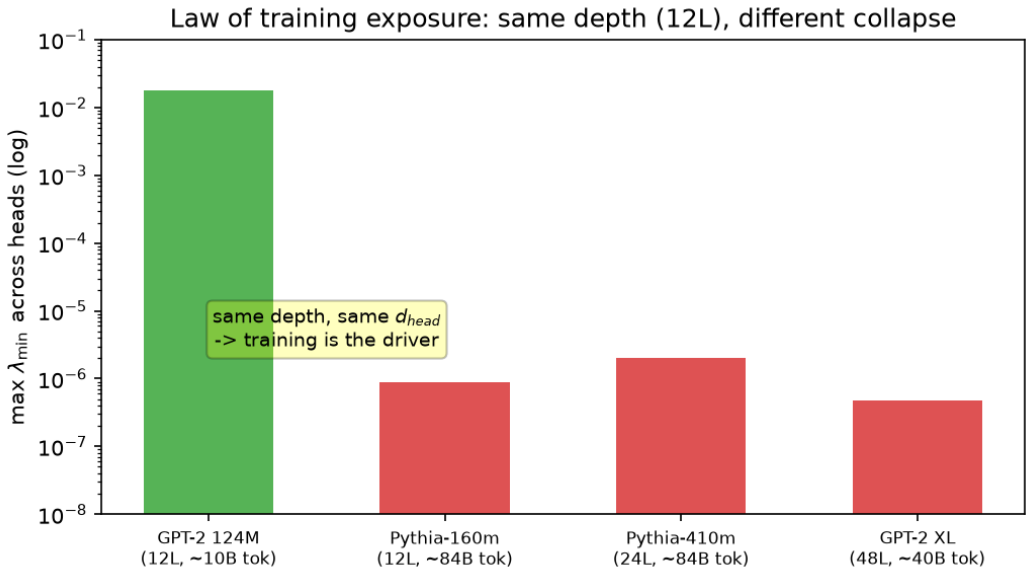


Figure 2: The natural experiment: Pythia-160m and GPT-2 124M share architectural depth (12 layers, $d_{\text{head}} = 64$) but differ in training exposure. Same depth, different collapse (Exp 37, 36).

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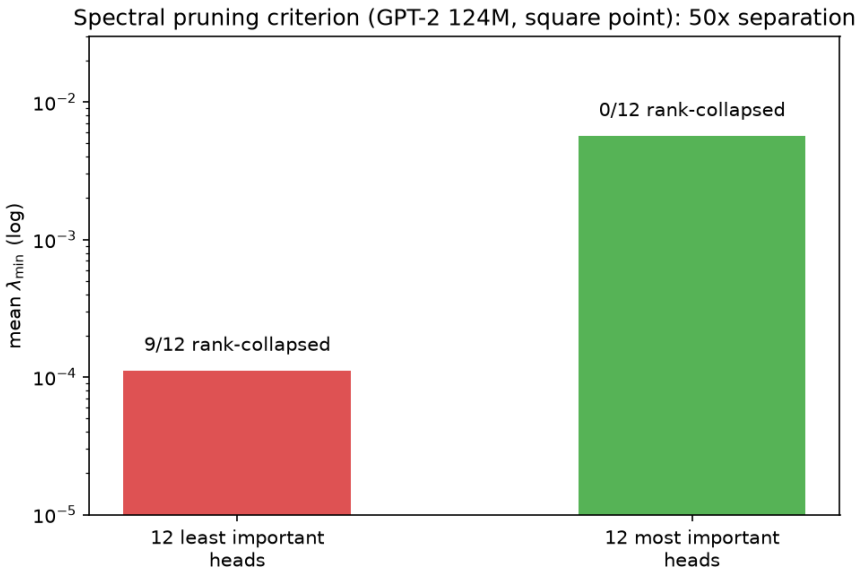


Figure 3: The spectral pruning criterion at the square point (Exp 32): the 12 least-important heads have 50× smaller mean $\lambda_{\min}$ and are 9/12 rank-collapsed; the 12 most-important are 0/12.

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